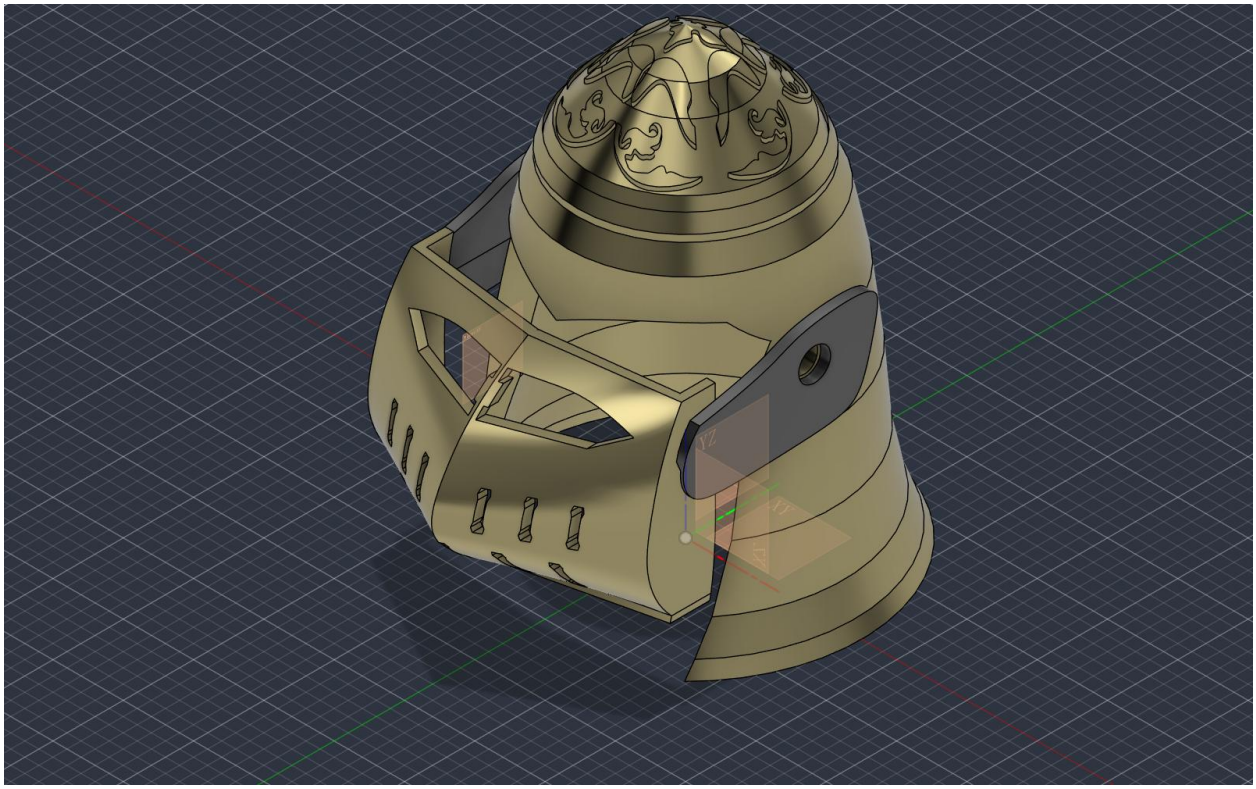
Ornate hinged helmet

Final project documentation | Student: Sam Shii | Source design: “final modeling”

This record describes the student’s final Fusion model and its actual timeline, including later changes to the shell, visor, hinge and decorative crown. The model is documented at the print-scale assembly state (history step 103); the final timeline state is an exploded print layout.

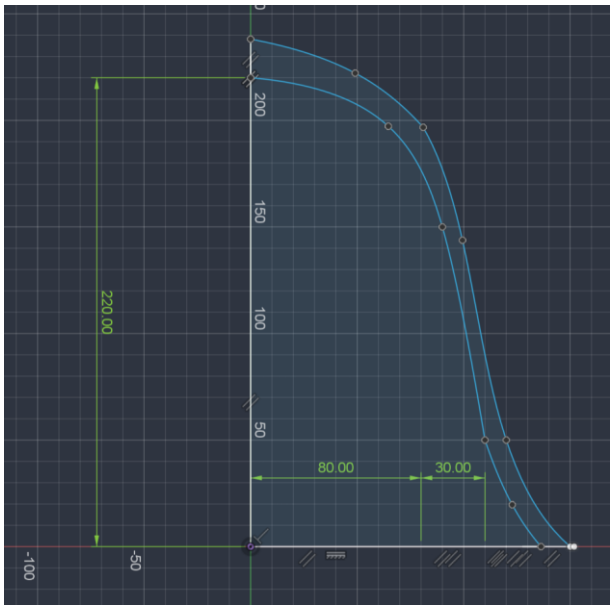


Assembly review in Fusion at history step 101, before the second scale operation. The source model is preserved in Final_Project_Original_Fusion_Model.f3d.

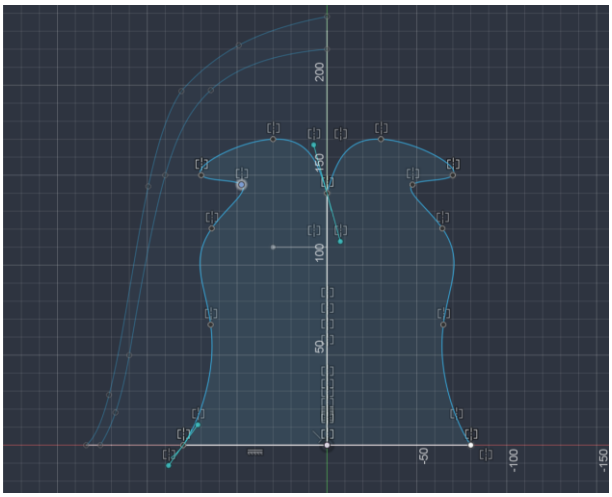
Evidence package: 111 numbered Fusion viewport captures; a 10-page contact sheet; the original Fusion archive; a separate Fusion drawing-source design; and a multi-sheet Fusion drawing.

1. Idea and design iterations

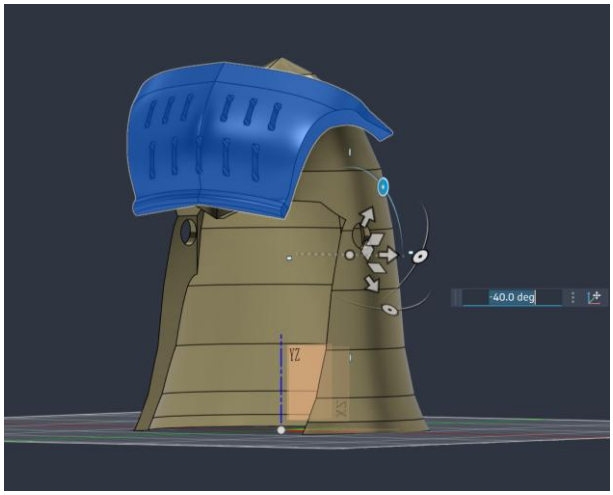
The initial idea was a medieval-inspired helmet with a pointed crown, long neck guard and pivoting visor. Early sketches compared shell profiles and visor shapes. The Week 4 ideation notes proposed a transparent multi-material version with internal decoration; the later CAD revision uses an ornate relief and gold-toned appearance. The visual appearance in Fusion is a concept finish, not proof of a physical material or post-process.



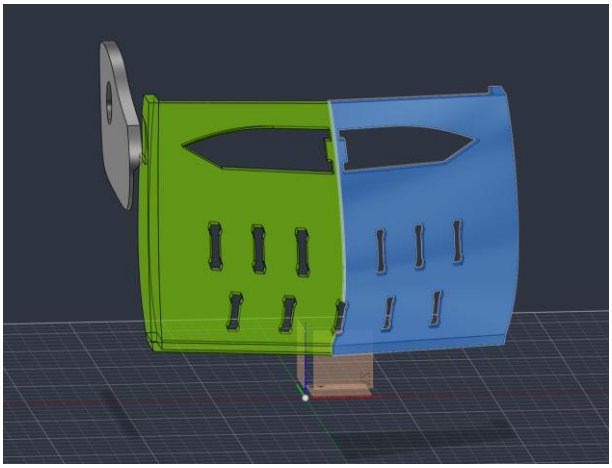
Side profile study: tall crown, curved outer wall and thickness.



Front study: central seam, eye-line position and symmetry.



Visor rotation was tested above the eye opening.

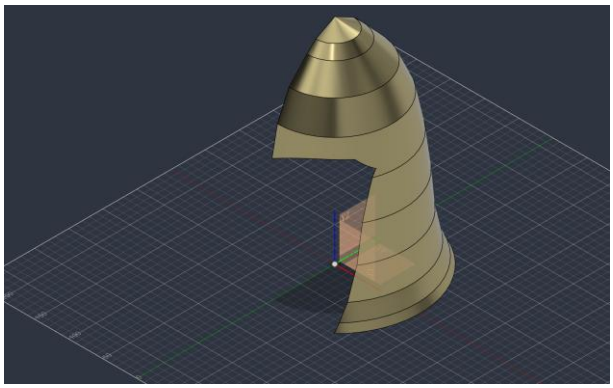


The visor was split into two sides with ventilation slots.

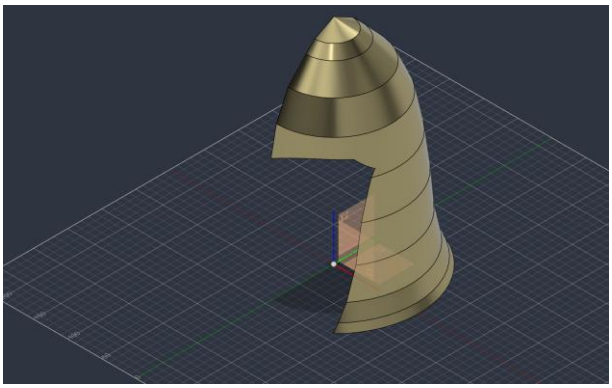
Design decisions visible in the final timeline: revolve the helmet wall; cut the face opening; split the shell; create and refine the hinged visor; add left and right hinge links; develop the crown pattern; then scale the assembly and separate the pieces for printing.

2. Shell: profile, opening and split

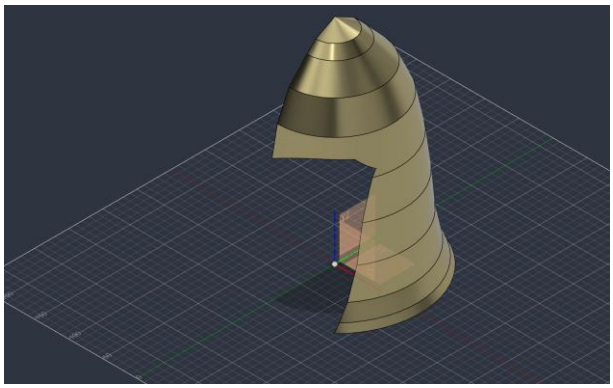
A sketched wall profile is revolved into the main shell. A cut opens the face area. A central split creates an accessible seam for a multi-part print and allows the shell shape to be investigated from both sides.



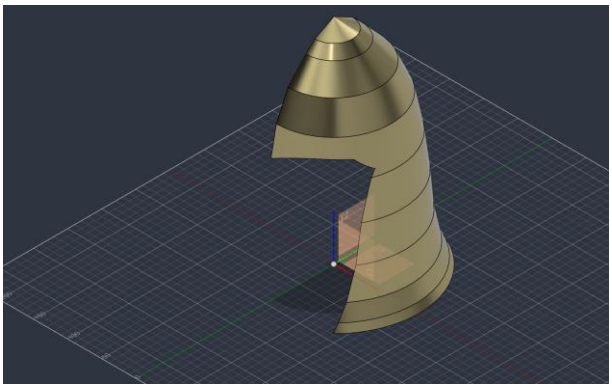
Step 011 — shell form after the revolve and face cut.



Step 017 — the open face and long neck wall remain visible.



Step 007 — Split1: the shell is separated on the center plane.

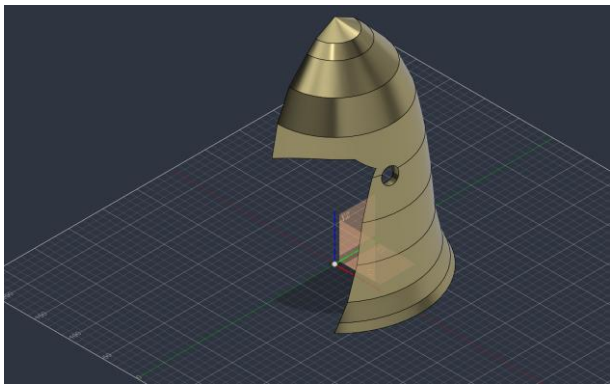


Step 015 — Loft2: early visor surface study.

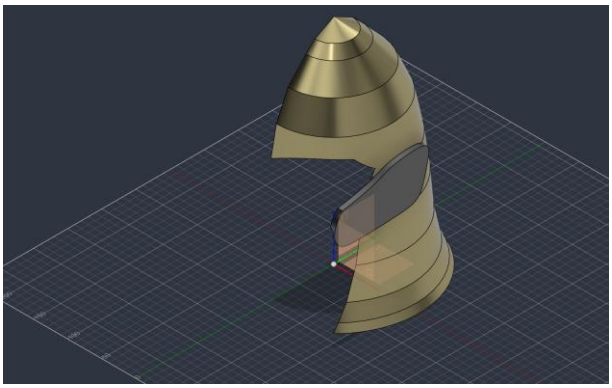
The split and the eye opening are important for printability: the open edges reduce enclosed supports, while the side seam gives a repeatable reference when fitting the shell and visor.

3. Visor and hinge development

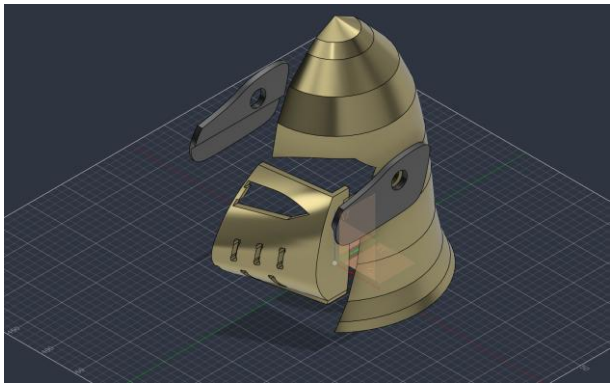
The visor profile was refined through sketches, extrusion, moves, splits and mirrors. The hinge links use separate bodies so the pivot motion can be inspected without merging the moving parts.



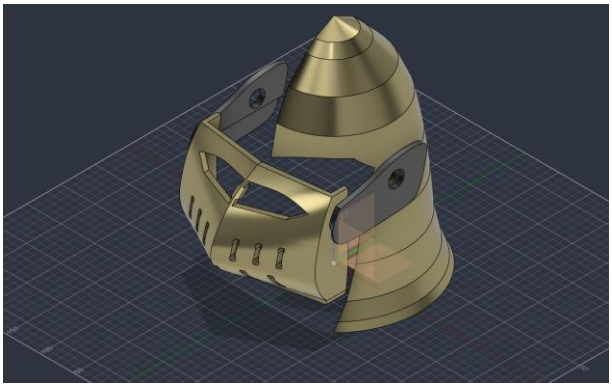
Step 051 — Extrude13: visor and side detail are developed.



Step 059 — Mirror1: opposite-side geometry is checked.



Step 065 — Move7: visor assembly study.



Step 077 — Mirror2: corresponding hinge/visor geometry is made.

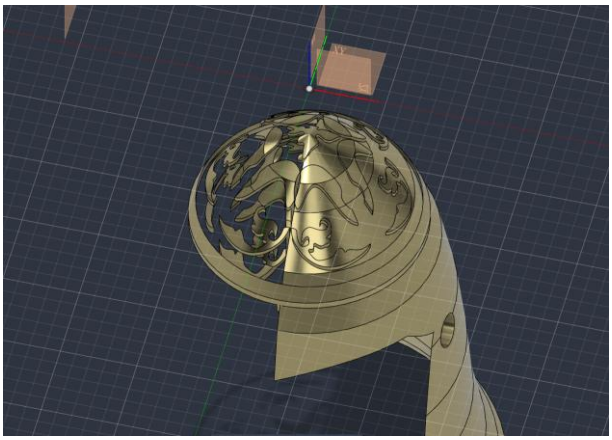
The final visible geometry contains four print bodies: one helmet shell, one visor and two hinge links. Separate STL exports exist for these four bodies. The two bolt meshes in the source folder are not final components and are excluded from the drawing set.

4. Decorative crown and final revisions

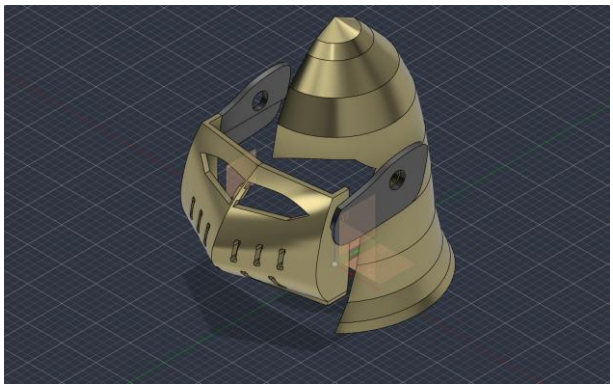
A reference canvas was used for the crown ornament. Later sketches and body operations integrate the motif on the top of the shell. The finished detail is best inspected at a larger viewport scale because fine features may be affected by the chosen print process.



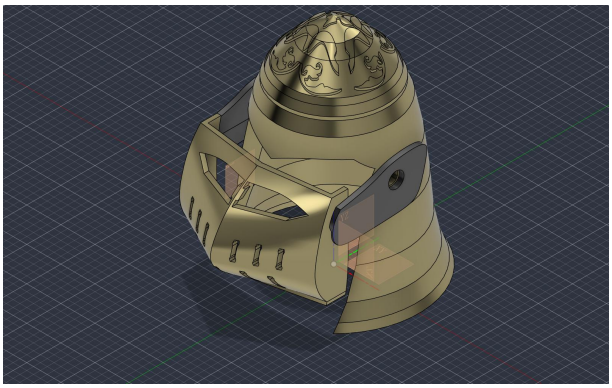
Decorative motif source retained in the student's final notes.



Student's Fusion capture showing the ornament on the crown.



Step 081 — reference canvas inserted.

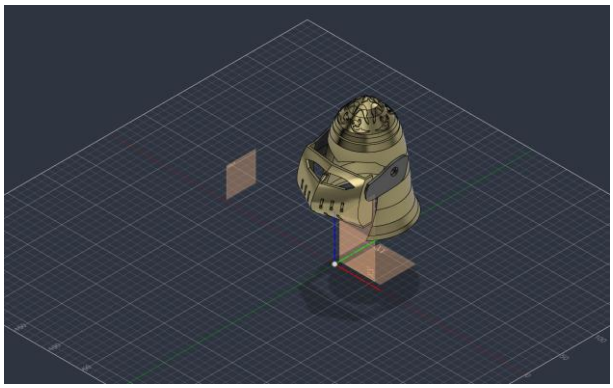


Step 101 — assembly after ornament and scale revision.

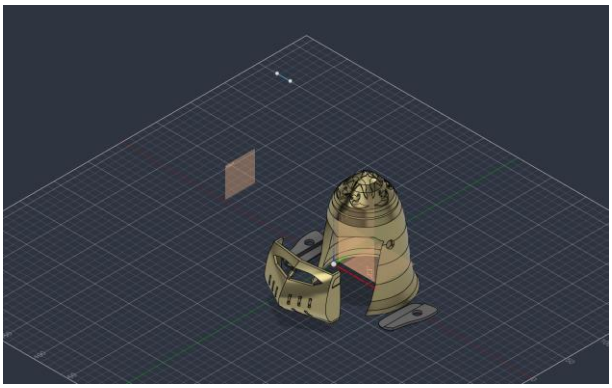
The motif is a visual element. The actual CAD export records a gold-toned appearance and a steel material assignment. These software settings do not establish the material of any fabricated print.

5. Scale, print layout and critical dimensions

The final scaling sequence changes the assembly size. The first scale stage reduces the assembly to 187.5 × 229.3 × 237.1. The second stage yields a print-scale assembled envelope of 112.5 × 137.6 × 142.3. Later Move features spread the four print bodies apart without defining a new assembled size.



Step 103 — print-scale assembly, 112.5 × 137.6 × 142.3.



Step 111 — separated final layout for export and printing.

Printed body	XYZ bounding box (mm)	CAD status
Helmet shell	112.50 × 97.65 × 142.27	Visible in final Fusion state
Visor	89.44 × 47.57 × 60.30	Visible in final Fusion state
Left hinge	32.52 × 62.23 × 3.15	Visible in final Fusion state
Right hinge	31.76 × 62.61 × 4.51	Visible in final Fusion state

Measurements are from Fusion body bounding boxes at the final history state. The assembled 100 mm requirement is evaluated at step 103, while the individual-body values above are for print preparation. Dimensions on the Fusion drawing should be checked against the selected process before fabrication.

6. Fabrication and finish record

The model is intended as a multi-part print. Print each visible body separately with a chosen orientation and supports appropriate to the final material. The shell opening, visor eye slit and vent slots must remain clear; inspect and test the hinge bores before committing to a full-scale print. The two small hinge links should be printed flat so their broad faces contact the build plate.

Material and post-processing status

Fabrication has not yet started. The student's Week 4 proposal identified clear resin for printed geometry and leather or EVA foam for a non-printed liner or strap; these remain material options. The final Fusion file shows steel material assignments with gold and satin appearance choices, which describe the CAD look only. The physical second material, sanding, polishing, painting and final assembly are planned rather than completed. Update this record with actual materials and finishing after fabrication.

Printability checks

- Exported helmet shell matches the visible Fusion body: 112.49 × 97.65 × 142.27 mm in the STL.
- The print-scale assembled envelope is 112.5 × 137.6 × 142.3 mm and clears the stated 100 mm threshold in all axes.
- The visor and hinge links remain separate solids; test pivot fit and clearance with a small section before full fabrication.
- Check thin crown relief, unsupported overhangs and ventilation slots in the slicer selected for the actual material.

Authorship note

The student's earlier ideation document discloses AI assistance for brainstorming and documentation. This record was assembled with AI assistance by reading the student's existing Fusion model and source notes. No physical print, finish or assembly has been completed.